An Interaction on a Flat Panel Display Using a Planar 1-DOF Electrostatic Actuator

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ABSTRACT
This paper demonstrates a new user-computer interaction on a flat panel display through physical motions of a paper sheet. A user handles a paper sheet directly by hand, and at the same time, a computer program drives the sheet using a planar electrostatic actuator. Thus a user and a computer can interact with each other through physical media. A simple program developed as a preliminary demonstration successfully demonstrated a new type of interaction.

ACM Classification: H.5.2 [Information Interfaces and presentation]: User Interfaces. – Interaction styles.

General terms: Design

Keywords: Interactive surface, physical tabletop interface, electrostatic actuator, sheet conveyer

INTRODUCTION
Recently, large-sized flat panel displays and embedded touch sensors have enabled intuitive tabletop interactions. In most cases, interactions are made by manipulating computer graphics with a help of touch panels. Instead of manipulating computer graphics, which do not *physically* exist, some studies have proposed to utilize real physical objects as interaction media [1-3]. In such systems, real objects for interaction are placed on tabletops, and users can operate computers intuitively through manipulation of those objects. Although the interaction is uni-directional in most of the studies, some studies attempted to realize bi-directional interactions using embedded actuators. In previously reported studies, electromagnetic motors have been utilized as embedded actuators to realize bi-directional interactions [2, 3]. However, electromagnetic motors have some difficulties to be integrated with flat panel displays because of their heavy and thick structures.

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An alternative solution has been proposed by using an electrostatic actuation principle [4]. In the proposed system, a large-scale planar electrostatic actuator was utilized that is characterized by its thin structure. Due to the thin structure, simply placing an actuator on a flat panel display can already realize integration of actuation functionality to interactive systems. The electrostatic actuator can directly handle dielectric sheets. In [4], however, actuated sheets need to have surface resistance around 10^{13}-10^{15} Ω, due to its actuation principle. This led to a need for specially treated sheet materials. It should also be mentioned that the work in [4] focused on operations of the electrostatic actuator, and did not demonstrate any user interactions.

In this paper, we utilize a similar, but different electrostatic actuation principle, in which normal PPC paper sheets can be directly used as interaction media. This paper also demonstrates, for the first time, a preliminary user-computer interaction using a planar electrostatic actuator. As a first step, we used an actuator in its simplest form. The employed actuator is opaque, has only 1-DOF for actuation, and does not cover the whole area of the flat panel display. In this work, considering such limitations, a simple demonstration program has been designed.

PLANAR ELECTROSTATIC ACTUATOR
The basic structure of the actuator utilized in this work is shown in Figure 1. The actuator consists of a thin plastic film and conductive three-phase electrodes embedded in the film. Although the actuator can be made transparent by us-

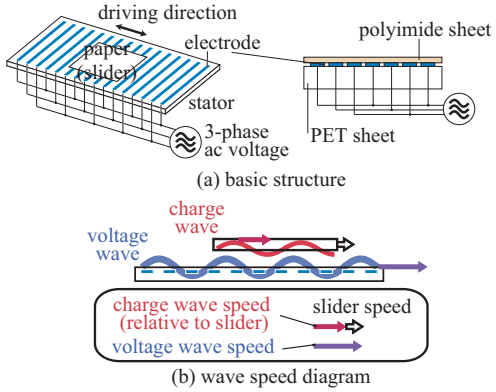


Figure 1 Structure and principle of the electrostatic actuator

ing transparent conductor, such as ITO, this work utilized opaque actuator due to a manufacturing reason.

The actuation principle, known as electrostatic induction actuation [5], is shown in Figure 1. Three-phase high voltage applied to the three-phase electrodes generates quasi-sinusoidal traveling field on the electrodes. The traveling field induces quasi-sinusoidal traveling charge distribution on the slider. Since the traveling speed of the charge distribution is lower than that of the exciting field, the sheet is electrostatically actuated to compensate for the speed difference. As reported in [5], the electrostatic actuation is not yet perfect; it sometimes shows unstable behavior. In this work, the unstable behavior was taken into account in designing a demonstration program.

PRELIMINARY DEMONSTRATION OF INTERACTION

Setup overview

Figure 2 shows the overview of the demonstration system. A 1-DOF opaque electrostatic actuator is placed on a LCD. Waveform generators and high voltage amplifiers are connected to drive the actuator. A USB-camera is installed above the LCD. For interactions, a PPC paper sheet cut into a small piece is placed on the actuator. The sheet is detected by a program through the USB-camera and its position is measured. The program renders some animations on the LCD, and at the same time, actuates the sheet on the actuator to follow what is happening in the animations. Simultaneously, a user can move the sheet by hand that invokes some interactive motions in the animation. In such a manner, a user can interact with the computer animation through the paper sheet.

Demonstration scenario: *interactive shadow play*

The developed simple demonstration program is *interactive shadow play* (see Figure 3). In the demonstration scenario, the electrostatic actuator represents a play area for shadow play. The play area is illuminated (only in the computer graphics) by a spotlight. The program monitors the play area using the USB-camera, and if it detects an object on the screen, the program renders its shadow on the screen for the shadow play. On the shadow play screen, there is a *shadow cat* that only exists as a shadow. The cat wanders around in the screen and if it finds a shadow of a paper sheet, it shows some interactive actions. For example, the

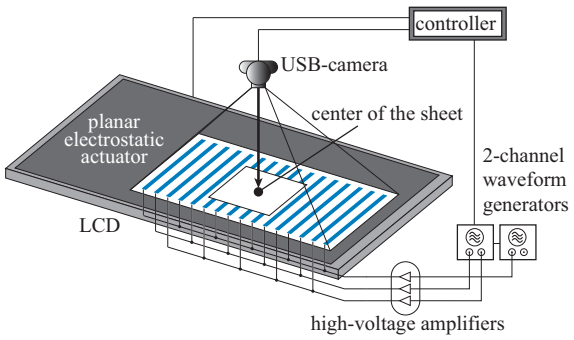


Figure 2 Setup overview

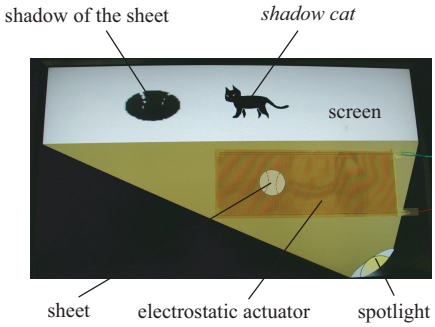


Figure 3 Snapshot of "interactive shadow play"

cat rushes to a sheet's shadow and pushes it that results in the real motion of the sheet. Pushing the sheet is realized by driving the electrostatic actuator. When a paper sheet does not move either due to the unstable behavior of the actuator or due to the user's action, the cat makes another action corresponding to the situation.

During these interactions, the shadow position and the sheet position are always synchronized; the shadow is always drawn on the corresponding position by the program. For a user, however, it seems as if the real sheet (on the actuator) follows the motion of the shadow (on the screen).

CONCLUSIONS

This paper demonstrated a preliminary user-computer interaction on a flat panel display using a planar electrostatic actuator. Our future studies include improvement of actuation stability, development of larger-size transparent actuators, and more dexterous operation using 2-DOF or 3-DOF actuators as mentioned in [5].

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